Chapter 29

Clinical Pharmacokinetics

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Learning Objectives:

Objective 29.1 Explain the basic concepts in clinical pharmacokinetics.
Objective 29.2 Discuss factors affecting pharmacokinetics of a drug.
Objective 29.3 Outline pharmacometrics.

OBJECTIVE 29.1. EXPLAIN THE BASIC CONCEPTS IN CLINICAL PHARMACOKINETICS

Clinical Pharmacokinetics and Terminology

Pharmacokinetics (PK) deals with the time course of a drug in the body following drug administration. Pharmacokinetic parameters describe various processes such as the absorption, distribution, metabolism, and excretion of a drug. Clinical PK is the application of PK principles for the safe and effective management of drug therapy in an individual patient. It is the blood that transports drugs to various organs in the body. The organs that clear the drug can eliminate it only if it is brought to that organ by the blood. Measurement of the time course of the drug concentrations in the blood is useful to determine the overall ability of the body to remove a drug. Pharmacokinetic principles help establish a relationship between drug concentrations measured in easily accessible body fluids (blood, plasma, urine, and saliva) and therapeutic effects and/or toxicities associated with a drug. For pharmacokinetic assessments, drug concentrations are typically measured in plasma or blood samples collected at various times after drug administration.

Drug concentration monitoring, e.g., in plasma is based on the concept of kinetic homogeneity. It means when plasma concentration of the drug increases, drug concentration in most of the other tissues increases proportionally. Same way when the concentration decreases in plasma, concentration in tissues also decrease proportionally. While pharmacokinetics describes the time course of absorption, distribution, metabolism, and excretion of drugs, pharmacodynamics relates drug concentration to drug response.

Routes of Drug Administration: Drugs are normally administered via different routes. Generally, the routes for drug administration are divided into two classes: intravascular route and extravascular route. When a drug is administered directly into the systemic circulation (blood circulation), it is called intravascular administration. On the other hand, when a drug is administered outside the vascular system, not directly into the systemic circulation, it is called extravascular drug administration. An extravascular route could be oral, sublingual, nasal, rectal, intramuscular, subcutaneous, or topical. If a drug is administered via extravascular route, it has to reach the blood, and the drug must cross various biological membranes to reach the systemic circulation before it travels to the site of action to elicit a pharmacological effect. Sometimes drugs can be administered directly at the site of action, called site-specific drug delivery. Site-specific drug delivery is e.g., used for local topical drug delivery and for local delivery of highly toxic drugs such as anticancer drugs to minimize unwanted/off target effects.

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All drugs undergo a series of processes after administration by different routes. Drugs are absorbed, distributed, metabolized, and finally excreted from the body. One of the basic concepts in pharmacokinetics is linear and nonlinear pharmacokinetics.

Linear Pharmacokinetics: If a drug follows linear PK (first order kinetics), the plasma drug concentration varies proportionately with the dose, but the pharmacokinetic parameters of that drug would remain constant, when the dose is changed. Dose normalized plasma concentration - time profiles will be superimposible. Most drugs that we currently use follow linear PK.

Nonlinear Pharmacokinetics: When plasma drug concentration changes disproportionate to the drug dose, and when a pharmacokinetic parameter varies depending on the administered dose, it is referred to as nonlinear PK. Nonlinear PK is the results of one or more saturable processes during drug absorption, distribution, metabolism, or elimination. Once elimination is saturated the quantity eliminated per unit time is same as this is the maximum ability of the body to eliminate the drug. For example, phenytoin follows linear PK at lower doses; however, at higher dose it follows nonlinear PK. See Table 29.1 for the differences between linear and nonlinear kinetics.

Absorption: Absorption is a process by which a drug molecule moves from the site of administration to the site of measurement (normally blood or plasma). The absorption can be a simple diffusion across a concentration gradient which is a passive process or involve a transporter or a specialized carrier which is an active process. Absorption normally starts immediately after drug administration and continues till the entire drug that can be absorbed is absorbed. Soon after drug administration, the absorption rate is much higher than the elimination rate. When the absorption rate is higher than the elimination rate, the drug level will increase in the body. At the time of maximal blood concentration (C_{max}), the absorption rate is equal to the elimination rate. After C_{max}, the elimination rate is higher than the absorption rate and the drug level will decrease in the body. Absorption is followed by distribution and/or elimination.

Distribution: Distribution is a reversible process of movement of drugs from and to the site of measurement. Distribution is influenced by multiple factors such as permeability of the drug in various tissues, blood flow, and binding of the drug to plasma or tissue proteins. In some cases, distribution of the drug is mediated by drug transporters. The distribution of a drug can be selectively altered by specific formulations such as liposomal formulations, nanodelivery systems, or targeted delivery systems such as antibody−drug conjugates.

Metabolism: Metabolism is a process of bioconversion of one chemical to another by various enzymes. A majority of the drugs undergo metabolism in the liver, but some drugs may be metabolized in the gut also. Other organs such as kidneys, lungs, and brain also contain drug metabolizing enzymes but their contribution to the overall metabolism of the drug is limited. Localized metabolism may affect the tissue levels of the drug and its metabolites, which may potentially influence the local concentration of a drug and its efficacy and/or toxicity.

Excretion: Excretion is the irreversible loss of unchanged drug from the body through bile, urine, sweat, etc. Elimination includes both metabolism and excretion. A fraction of the dose administered can be excreted unchanged (fe) via kidney and the remaining fraction of the dose may undergo metabolism (1 − fe). Drugs mainly excreted unchanged through urine like metformin have a high fe, whereas drugs like amiodarone that are mainly metabolized by liver have a high fm (1-fe).[1–4]

TABLE 29.1 Linear and Nonlinear Kinetics

Linear Pharmacokinetics	Nonlinear Pharmacokinetics
Pharmacokinetics parameters are dose independent	Pharmacokinetics parameters are dose dependent
Follows first order kinetics	Follows mixed order kinetics
Does not involve saturable process	Saturable processes are involved
Plasma concentration−time profiles are superimposable at wide dose ranges, when dose normalized	Plasma concentration−time profiles are not superimposable at wide dose ranges, when dose normalized

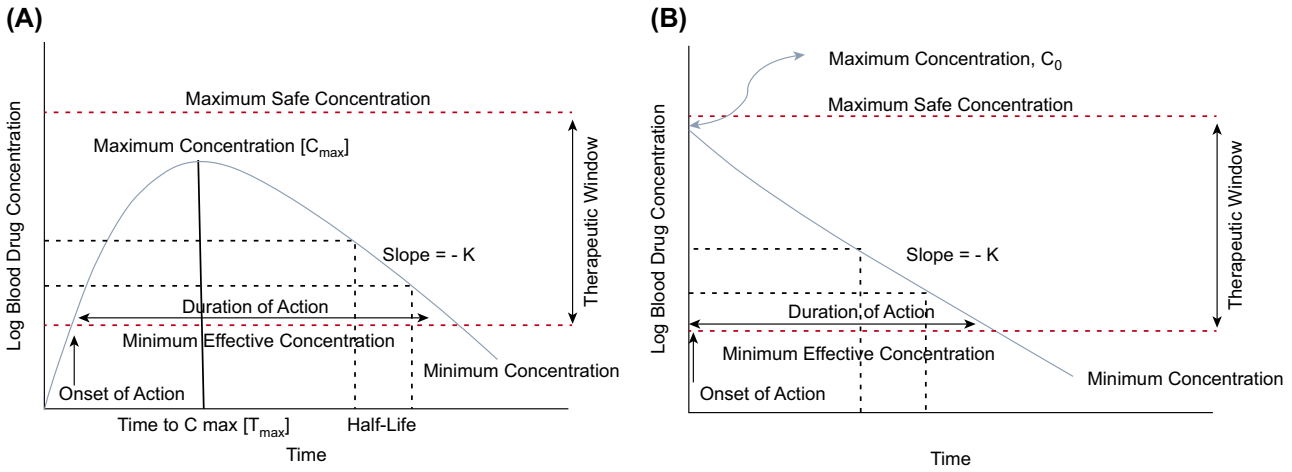


FIGURE 29.1 Blood/plasma drug concentration versus time profile of a drug after oral (A) and IV (B) administration.

Fig. 29.1 represents a typical plasma concentration versus time profile of a drug after oral (A) and IV (B) administration. Information about the PK parameters, T_{max}, C_{max}, AUC, MIC, MEC, and TW can be visualized.

Eqs. (29.1) and (29.2) can be used to describe the plasma concentrations of a drug after oral and intravenous single dose administration, respectively, at any time (t) after drug administration.

$$C = \frac{F \cdot D \cdot k_a}{V_d(k_a - k_e)} \times (e^{-k_e \cdot t} - e^{-k_a \cdot t}) \tag{29.1}$$

plasma concentration of a drug after oral administration

$$C = C_0 \times e^{-K_e t} \tag{29.2}$$

plasma concentration of a drug after IV dose

here, D = dose; F = fraction absorbed; C = plasma concentration at time t; C_0 = plasma concentration at time 0 (immediately after IV administration); V_d = volume of distribution; K_a = absorption rate constant; K_e = elimination rate constant; and $e^{-Ke \cdot t}$ = fraction or percent of the drug that remains in the body at a certain time after dose (t). This means $1 - e^{-Ke \cdot t}$ is the pecent or fraction of the drug excreted after time t.

Maximum Concentration: Maximum concentration (C_{max}) is the highest concentration of the drug observed within a dosing interval. It is also called the peak concentration (C_{peak}). After IV bolus administration, $C_{max} = C_0$ (g/L), where C_0 is the drug concentration immediately after the completion of IV bolus dose. After oral administration, C_{max} can be observed from the concentration–time profile and it can be estimated using the following equation:

$$C_{max} = \frac{F \times D}{V_d} e^{-K_e T_{max}} (g/L) \tag{29.3}$$

where, F is the fraction of drug absorbed, D is the dose, V_d is the volume of distribution, K_e is the elimination rate constant, and T_{max} is the time to C_{max} (Fig. 29.1A).

Minimum Concentration: Minimum Concentration (C_{min}) is the lowest concentration of a drug within a dosing interval. It is also called trough concentration (C_{trough}), when drugs are administered repeatedly. The value of C_{min} can be obtained from the concentration–time profile after IV administration (Fig. 29.1B) and can be calculated using the following equation:

$$C_{min} = C_0 e^{-K_e T_{last}} (g/L) \tag{29.4}$$

where, C_0 is maximum concentration, K_e is elimination rate constant, and T_{last} is the time of the last blood sample collected within a dosing interval (Fig. 29.1B).

Time to Maximum Concentration: Time to Maximum Concentration (T_{max}) is the time taken for the drug concentration to achieve its highest value within a dosing interval. The value of T_{max} can be observed from the concentration−time profile and it can be calculated using the following equation:

$$T_{max} = \frac{\ln \frac{K_a}{K_e}}{(K_a - K_e)} (hr) \tag{29.5}$$

where, T_{max} is the time taken to reach maximum concentration, K_a is the absorption rate constant, and K_e is the elimination rate constant.

Minimum Effective Concentration: Minimum effective concentration (MEC) is the minimum concentration needed to elicit desired therapeutic effect. It is expressed as gm/L (Fig. 29.1A and B).

Maximum Safe (Tolerable) Concentration: Maximum safe concentration (MSC) is the maximum concentration with no undesired effects and beyond which toxicity is observed. It is also called minimum toxic concentration. It is expressed as g/L.

Therapeutic Window: Therapeutic window (TW) can be defined as the range of drug concentrations that provide therapeutic response without significant adverse effects. Drug concentrations above MSC are associated with toxicity, and the drug concentration below MEC does not provide adequate therapeutic effect leading to therapeutic failure. The ratio between MSC and MEC is called the therapeutic index (TI). The TI is a measure of the safety of the drug. The larger the TI, the safer the drug.

Half-Life: Half-life ($T_{1/2}$) is the time taken for the drug amount or concentration to decline to half of its original value. Most of the drugs follow first order kinetics. Mathematically, it is calculated using the following equation:

$$T_{1/2} = 0.693/K_e \text{ or } 0.693 \times V_d/CL \tag{29.6}$$

where, 0.693 is the natural log of 0.5 meaning half, K_e is the elimination rate constant, CL is the clearance, and V_d is the apparent volume of distribution. $T_{1/2}$ is typically represented in minutes to hours. Half-life is useful to determine how long it will take the body to eliminate the entire drug, to determine the duration of drug action, and to determine the dosing interval and the time to reach steady state for medications administered chronically.

Elimination Rate Constant: Elimination rate constant (K_e) describes drug loss by all elimination processes and the unit of elimination rate constant is per unit time (sec^{-1}, min^{-1}, h^{-1}). Mathematically, K_e can be calculated from the slope of the terminal disposition phase of the ln plasma concentration versus time profile and by the equation

$$K_e = [\ln(C_1) - \ln(C_2)]/(T_2 - T_1) \tag{29.7}$$

where, C_1 and C_2 are plasma/blood concentrations at time T_1 and T_2, respectively.

Area Under the Curve: The area under the curve (AUC) represents total systemic drug exposure after a dose. The AUC is calculated from the concentration−time profile data by the trapezoidal rule, where AUC is divided into discrete set of rectangles. Each rectangle represents the area for that interval and the sum of area of trapezoids represents AUC up to the last sampling time. The AUC from last sampling time point to ∞ is calculated as shown below and the unit of AUC is amount/volume/time; ng/mL/h. AUC is calculated using the following equation:

$$AUC_{0-\infty} = \sum \frac{(C_n + C_{n+1}) \times (T_{n+1} - T_n)}{2} + \frac{C_{last}}{K_e} \tag{29.8}$$

Clearance: Clearance is the term used to describe the ability of the body to eliminate a drug from the body. The unit of clearance is the volume of blood or plasma that is cleared of the drug per unit time (mL/h, mL/min, L/h, etc.). Clearance is normally described as blood clearance, plasma clearance (depending on where the drug concentration is measured), and may be based on total drug concentrations or free (unbound) drug concentrations. For example, clearance of phenytoin is 10 L/h, and this means that in 1 h, an equivalent of 10 L of blood gets cleared of phenytoin (amount of the drug removed from body is equivalent to what is present in 10 L of blood).

Normally, more than one organ is involved in the elimination of drugs. Although the most important organ that is involved in drug elimination is the liver, kidney also contributes to the removal of many drugs. Additionally, lung and intestine can also contribute to the total clearance of a drug. Clearance is an additive process that includes contribution

from all the organs. Total body clearance of a drug is the sum of hepatic clearance, renal clearance, and clearance by other organs.

$$CL_{total} = CL_h + CL_r + CL_{other} \tag{29.9}$$

where, CL_{total} = total body clearance, CL_h = hepatic clearance, CL_r = renal clearance, and CL_{other} = clearance by other organs.

Clearance of a drug by a particular organ depends on the blood flow to the organ and the ability of the organ to extract the drug. Extraction ratio (E) is a term used to describe fraction of drug being removed by a specific organ during a single passage. ER is measured from the concentration of drug in the incoming artery (C_a) and the concentration of drug in the outgoing vein (C_v) leaving that organ and with the organ blood flow (Q). ER is dependent on the hepatic blood flow (Q), intrinsic clearance (CL_{int}), and fraction of the drug unbound in plasma (f_u).

$$E = \frac{Q \times (C_a - C_v)}{Q \times C_a} = \frac{C_a - C_v}{C_a} \tag{29.10}$$

$$CL_{organ} = Q_{organ} \times \frac{(C_a - C_v)}{C_a} = Q_{organ} \times E = (Q \times f_u\ CL_{int}) / (Q + f_u\ CL_{int}) \tag{29.11}$$

A drug is considered to be a high extraction ratio drug if E is greater than 0.7, on the other hand if E value is less than 0.3, the drug is considered a low extraction ratio drug. If E is between 0.3 and 0.7, the drug is considered intermediate extraction ratio drug. Lidocaine and propronolol are examples of drugs with high hepatic extraction ratio. As most of the drug is cleared in one passage through liver, clearance is depended on hepatic blood flow. From Eq. (29.11), also it is clear that if the entire drug is being removed by a particular organ during single pass (high extraction ratio), CL equals to the hepatic blood flow. Clearance of a drug can be estimated from area under plasma concentration time curve and dose by using Eqs. (29.12) and (29.13).

$$CL = \frac{Dose}{AUC}, \text{for IV adminstration} \tag{29.12}$$

$$CL = \frac{F\ Dose}{AUC}, \text{for extravascular administration} \tag{29.13}$$

where, F is the fraction bioavailable. Typically, when the drug is given by extravascular route, the value of F is unknown and the apparent clearance calculated as dose/AUC is CL/F.

Volume of Distribution: Volume of distribution (V_d) defines the volume of fluid which would be required to contain the amount of drug in the body based on plasma concentration measurements. The volume of distribution does not represent a true physical volume, rather it is a hypothetical volume which is different than the actual human body volume. V_d does not provide any indication about the types of tissue or organ into which drug has accumulated or distributed.

$$V_d = \frac{\text{Amount of Drug at time 0}}{\text{Plasma drug concentration at time 0}} \ldots \text{Dose}/C_p^{\,0}, \text{ where } C_p^{\,0} \text{ is the concentration at time zero} \tag{29.14}$$

When a drug is administered into the blood, depending on the characteristics, a certain portion of the drug binds to the plasma proteins (albumin, α-1-acid glycoprotein, or blood cells) and the remaining drug stays as free drug in the plasma. It is only the free drug, which is not bound to the blood cells or plasma proteins, which can cross cell membrane and distribute into the other organs or tissues. So, if a drug is 100% bound to plasma proteins, the volume of distribution will be equal to the plasma volume (3 L) or if certain portion binds to plasma and blood cells, the volume of distribution might be equal to total blood volume (5.5 L). On the other hand, if some of the drug is free in the plasma and distributes to the tissue but remains only in the extracellular water, the volume of distribution might be equal to total water content in the body (42 L). If a drug is highly taken up in the tissues, V_d values could be a large number. For example, digoxin (a cardiac drug) is highly distributed into tissues and has a volume of distribution of about 400 L, which is almost 10 times of the total human body volume (70 kg human).[5] Table 29.2 represents the commonly used medications with their clearances class, volume of distribution, total clearance, half-life, and route of administration. Total volume of distribution is physiologically related as follows:

$$V_d = V_p + V_{tw} \times \frac{f_u}{f_{ut}} \tag{29.15}$$

here, V_p = volume of plasma, V_{tw} = volume of water outside the plasma, f_u = free fraction of the drug in plasma, and f_{ut} = unbound fraction of drug in the tissue.

TABLE 29.2 Route of Administration, Clearances, Volume of Distribution and Half-Life of Top 15 Drugs Prescribed in the United States (2017)

Drug	Route of administration	Clearance CL or CL/F[a]	Volume of distribution Vd (or) Vd/F[a]	Half-life (t1/2)
Lisinopril	PO	3 L/h	0.5 L/Kg [b]	12 h
Levothyroxine	PO	11.8 ± 5.5 L/h	0.043 L/Kg [b]	6–7 days
Atorvastatin	PO	37.5 L/h	5.44 L/Kg [b]	14 h
Metformin HCL	PO	36 ± 7.92 L/h	9.34 ± 5.11 L/Kg [b]	6.2 h
Simvastatin	PO	52.59 L/h	67.5 L/Kg [b]	5.8 h
Omeprazole	PO, IV	30–36 L/h	0.3 L/Kg	0.5–1 h
Amlodipine Besylate	PO	40 L/h	21 L/Kg	30–50 h
Metoprolol	PO, IV	0.05L/h	3.2–5.6 L/Kg	3–4 h
Acetaminophen	PO	19 L/h	0.8 L/kg	2.4 h
Albuterol	Inhaler	263 L/h	9 ± 2.4 L/Kg [b]	6 h
Hydrochlorothiazide	PO	21 L/h	4.19 L/Kg	5.6–14.8 h
Losartan Potassium	PO	36 L/h	0.5 L/Kg [b]	2 h
Gabapentin	PO	11.4 L/h	0.83 L/Kg [b]	5–7 h
Sertraline HCL	PO	96 L/h	20 L/Kg	26 h
Furosemide	PO, IV	11.64 L/h	0.2 L/Kg [b]	2 h

[a] *Vd/F and CL/F may be high due to low F.*
[b] *Values converted based on 70 kg body weight.*

Multiple Dosing: Drugs are administered to achieve a desired plasma concentration within the TW. In some disease conditions, drug therapy continues for a long period of time or even lifelong, such as the case with antidiabetic or antihypertensive drugs. For these chronic conditions, multiple dosing of the drug is required. In case of multiple doses, accumulation of the administered drug occurs depending on the half-life and dosing interval until steady-state plasma concentration is achieved, where the rate of input equals to the rate of output. For example, assume a drug has a half-life of 8 h. If a second dose is administered 8 h after administration of the first dose, 50% of the previous dose will still remain in the body. Whereas, if we administer the second dose 16 h after administration of the first dose, only 25% of the previous dose will remain in the body. So, accumulation is dependent on drug half-life and dosing interval and calculated by the following equation:

$$R_{acc} = \frac{1}{1 - e^{-K\tau}} \tag{29.16}$$

where, R_{acc} = accumulation ratio, K = elimination rate constant, and τ = dosing interval. If the accumulation ratio is 1, that means there is no accumulation. On the other hand, $R_{acc} > 1$ indicates accumulation. At steady state, there will be no accumulation because the drug input is equal to the drug output. The time to reach steady state depends on the half-life of the drug, such that 3.3 half-life is required to achieve 90% steady state and 5 half-life is required to achieve 97% steady state. In case of multiple dosing, steady-state plasma concentration does not mean a fixed concentration of drug is present in the plasma. At steady state, plasma level fluctuates between maximum $\left(C^{\infty}_{max}\right)$ and minimum $\left(C^{\infty}_{min}\right)$ plasma concentrations and the plasma concentration time profile will be identical from one dose to the next. The relationship between $\left(C^{\infty}_{max}\right)$ and $\left(C^{\infty}_{min}\right)$ is as follows:

$$C^{\infty}_{min} = C^{\infty}_{max} \times e^{-K\tau} \tag{29.17}$$

One important PK parameter at steady state is the average steady-state concentration $\left(C^{\infty}_{avg}\right)$, though $\left(C^{\infty}_{avg}\right)$ does not give any indication about the degree of fluctuations. Eq. (29.18) dictates the relationship among $\left(C^{\infty}_{avg}\right)$, $\left(C^{\infty}_{min}\right)$, and $\left(C^{\infty}_{max}\right)$ as follows:

$$C^{\infty}_{avg} = \frac{C^{\infty}_{max} - C^{\infty}_{min}}{\ln \frac{C^{\infty}_{max}}{C^{\infty}_{min}}} \tag{29.18}$$

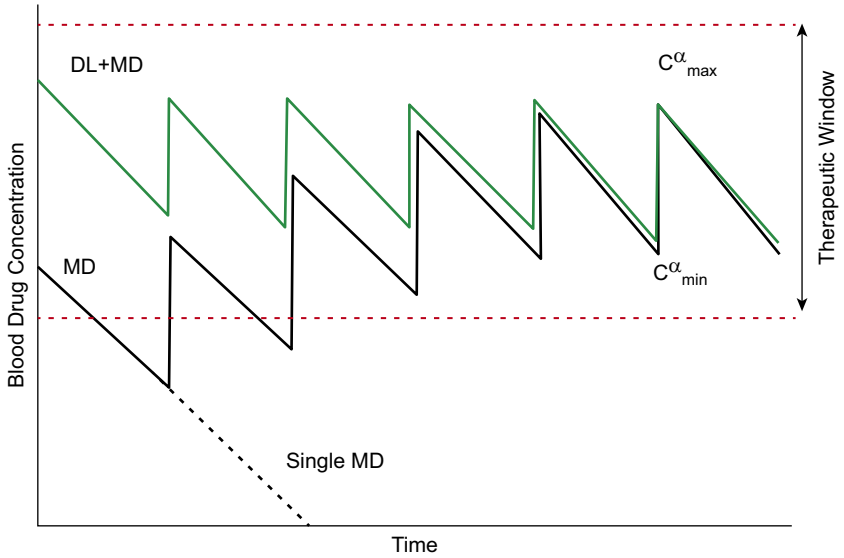
In case of oral dosing, $\left(C^{\infty}_{avg}\right)$ could be calculated from dose (D), fraction bioavailable (F), clearance (CL), and dosing interval (τ) as indicated by Eq. (29.19)

$$C^{\infty}_{avg} = \frac{F\left(\frac{D}{\tau}\right)}{CL} = AUC/\tau \tag{29.19}$$

In clinical practice, we want to achieve steady state as early as possible, and minimize the fluctuations between minimum and maximum drug concentrations. Once a steady state plasma concentration is achieved but if a patient is not responding to the drug, increasing the dose or decreasing the dosing interval or both together are options to increase plasma concentration of the drug. Once a dose is changed, it will take approximately 4–5 half lives to achieve new steady state plasma concentration. If a drug has longer half-life, more time will be required to achieve desired steady state as compared with a drug with shorter half-life. For example, half-life of phenobarbital is 4 days, so 20 days would be required to achieve a steady state (5 half lives). To avoid this delay in achieving steady state, maintenance dose is preceded by a loading dose to achieve and maintain steady state quickly.

Loading Dose: A loading dose is the initial dose of the drug that is given to achieve target plasma concentration faster. Fig. 29.2 illustrates the blood concentration vs time curve after multiple IV dose of a drug (MD) without a loading dose, and with a loading dose (LD). As shown in the figure, a loading dose is given to reach the desired concentration faster.

Maintenance Dose: Maintenance dose is the dose of drug that is to be administered to keep the plasma concentration of a drug within a certain level at steady state. When we only administer a maintenance dose, it will take a long time to



DL = Loading Dose, MD = Maintenance Dose

FIGURE 29.2 Plasma concentration–time profile after loading and maintenance dose.

reach steady state. Whereas, when we provide a loading dose followed by maintenance dose, desired plasma level is achieved earlier. Fig. 29.2 shows the scenario of loading dose with maintenance dose for IV bolus injection. The relationship between loading dose and maintenance dose is expressed by following equation:

$$D_L = D_M \times R_{acc} \tag{29.20}$$

Additionally, required loading dose can be calculated from desired maximum plasma concentration at steady state $\left(C^{\infty}_{max}\right)$ and volume of distribution

$$D_L = V \times C^{\infty\,max} \tag{29.21}$$

Pharmacokinetic Models

Movement of the drug within the body and out of the body is a very complex process because a drug passes through a series of biological processes before it enters or exits from the body. The major processes which a drug goes through in the body are absorption, distribution, metabolism, and elimination. The best way to explain drug behavior within the body is with the use of mathematical models and equations. The most commonly used models for pharmacokinetic estimations are referred to as compartmental models. Compartmental models describe the movement of a drug within and out of the body. There are one, two, and multicompartment models. Each compartment does not necessarily represent any specific organ or tissue, rather they represent a group of tissues or organs with similar time course of drug concentrations. The compartmentality of a drug is determined by the number of compartments required to explain the plasma concentration−time profile of that particular drug. Imagine an example, where a drug X easily distributes into blood, liver, and kidney, but distribution of same drug (X) into the brain happens very slowly. We can combine blood, liver, and kidney into one compartment, but the brain will be in a separate compartment. The compartment that includes blood and highly perfused organs (such as liver, lung, and kidney) is called the central compartment. And compartment that includes slowly perfused organs or tissues (brain, adipose tissue) is called the peripheral compartment.

One Compartment Model

In case of a drug that can be described by a one compartment model, the entire body is considered as a single unit and the drug instantaneously distributes throughout the body. In a one compartment model, drug irreversibly exits from the

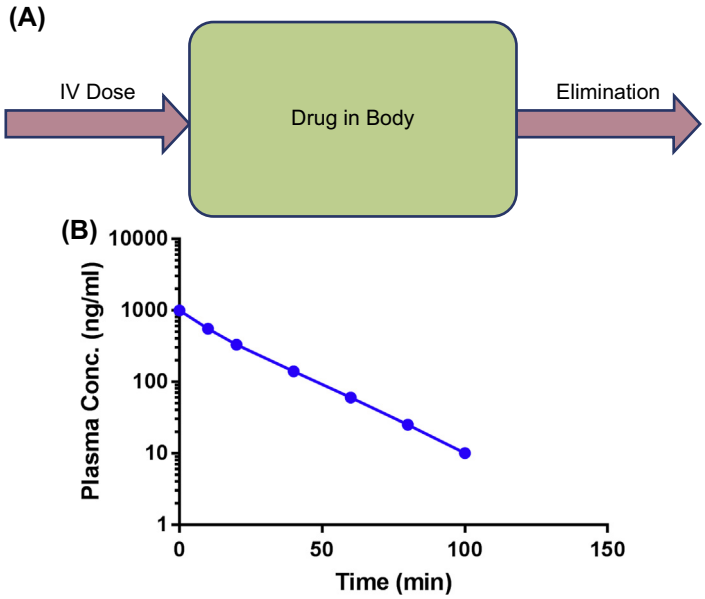


FIGURE 29.3 (A and B) One compartment model.

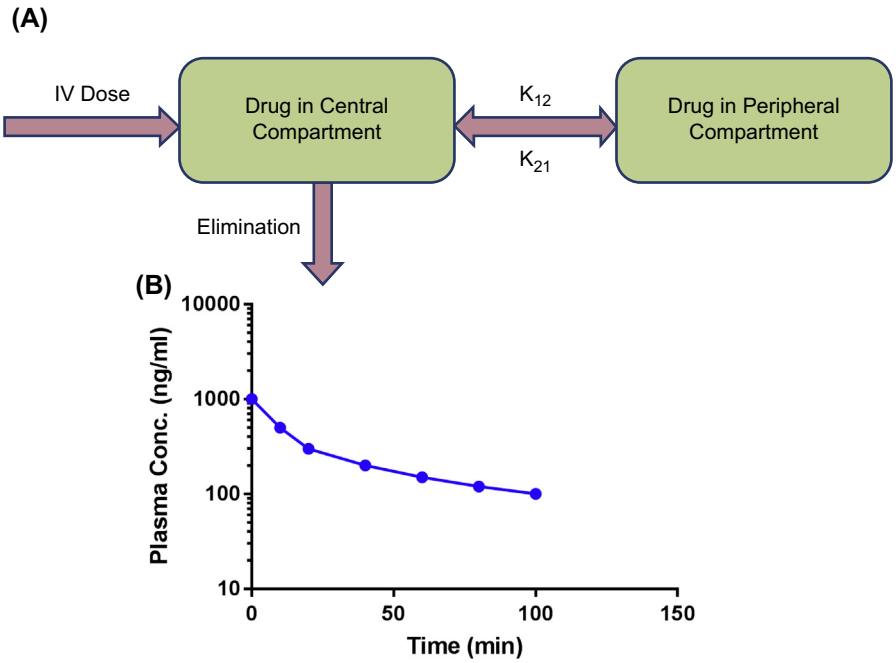


FIGURE 29.4 (A and B) Two compartment model.

compartment (Fig. 29.3A). The plasma concentration versus time profile on a semilog plot shows a straight line if drug follows one compartment model (Fig. 29.3B).

Fig. 29.3A represents a one compartment model. Drug enters into the body from administration site. Drug is irreversibly eliminated from the body. Additionally, Fig. 29.3B represents plasma concentration time profile a drug following one compartment model in semilog graph after IV administration.

Two Compartment Model

Fig. 29.4A represents a two compartment model. Drug enters the central compartment from administration site. Drug is either irreversibly eliminated or distributed into the peripheral compartment. The drug in peripheral compartment can redistribute to central compartment; K_{12} and K_{21} are the rate constants from central to peripheral and peripheral to central compartment, respectively. Fig. 29.4B represents a plasma concentration time profile of drug following two compartment model on a semilog graph after IV administration, a curvilinear profile showing distribution followed by elimination. E.g., digoxin after IV administration exhibits a rapid decline in plasma due to distribution to cardiac tissue. Digoxin has high affinity to skeletal and cardiac tissues compared to plasma. The cardiac tissue to plasma digoxin concentration ratio is 70:1.

OBJECTIVE 29.2. DISCUSS FACTORS AFFECTING PHARMACOKINETICS OF A DRUG

Absorption

Several factors can influence the rate and extent of absorption of drugs. These can be categorized as physicochemical, physiological, and formulation factors. Physicochemical factors include solubility in aqueous system, lipophilicity, size and structure, ability to form hydrogen bonds, polar surface area, chemical stability, and susceptibility to enzymatic reactions. Physiologic factors include gastric pH, gastric motility, intestinal motility, fluid volume, surface area, food–drug interactions, drug–drug interactions, intestinal metabolism, and intestinal uptake and efflux transporters. Formulation factors include disintegration and dissolution/release of drug from dosage form.

The biopharmaceutics classification system (BCS) classifies drugs based on their permeability and aqueous solubility into four classes. Class I, drugs are highly soluble and highly permeable. These drugs are normally completely absorbed

from GI tract and show physicochemical properties for optimum oral bioavailability. Normally, dissolution from solid dosage form is the rate-limiting step in the overall concentration time profile of Class I drugs. The overall bioavailability of these drugs can be affected by chemical degradation, complexation, or presystemic/first pass metabolism. The in vivo performance of a dosage form of a Class 1 drug can be readily assessed in vitro, leading to potential biowaiver. Acetaminophen, piroxicam, midazolam, and propranolol hydrochloride are drugs that are highly water soluble and have high permeability. They are well absorbed from the GI tract and are classified as Class I drugs.

Class 2 drugs show low solubility, but high permeability. These drugs show rapid absorption from solution or from a readily absorbable dosage form. Formulation strategies can help in overcoming the lower aqueous solubility of the drugs and in improving the dissolution rate of the product. The in vivo performance of a dosage form of a Class 2 drug can be readily assessed in vitro, leading to potential biowaiver. Most of the immunosuppressive drugs such as tacrolimus, sirolimus, mycophenolic acid, and cyclosporine fall into this category. Drugs in this class show moderately rapid absorption ($T_{max} \sim 1-4$ h). Improved and consistent absorption of cyclosporine from the microemulsion compared with olive oil formulation and tacrolimus extended release formulation compared with immediate release tablet that demonstrated the significance of the formulation in the improved PK of Class 2 drugs. Similarly, lipid-lowering agents such as simvastatin, rosuvastatin, and other statin drugs belong to BCS Class 2.

Class 3 drugs show high aqueous solubility and low permeability. Intestinal permeability is the rate-limiting step in the absorption of these drugs. Class 4 consists of poorly soluble and poorly permeable drugs. These drugs are poorly bioavailable after oral administration and are typically used for local action.

A recently proposed Biopharmaceutics Drug Disposition Classification System (BDDCS) is a complimentary system to BCS that predicts the importance of transporters in determining drug disposition, as well as in predicting drug–drug interactions. Both BCS and BDDCS use solubility as one of the two classification criteria, but in BDDCS, predictions are based on extent of drug metabolism. Class 1 drugs are highly soluble and highly metabolized; Class 2 drugs are poorly soluble and highly metabolized indicating involvement of efflux transporter in gut and both uptake and efflux transporters in the liver; Class 3 drugs are highly soluble and poorly metabolized predicting involvement of uptake transporters; and Class 4 drugs are poorly soluble and poorly metabolized predicting significance of uptake and efflux transporters. Class 2 drugs such as tacrolimus, sirolimus, and cyclosporine were predicted to have predominant effect of efflux transporters, and the experimental results have proven these predictions to be true.

The overall systemic availability of a drug after oral administration can be calculated by the following equation:

$$F = f_a \times f_g \times f_h \tag{29.22}$$

where f_a is the fraction of the drug absorbed, f_g is the fraction of the drug going through gut and reaching portal circulation, and f_h is the fraction of drug passing through liver during first pass and reaching systemic circulation.

f_a will be less than one if the drug is either incompletely absorbed or degraded/metabolized or effluxed while passing through the gut. F_g will be less than 1 if a drug is metabolized in the gut wall. F_h will be less than 1 if the drug is metabolized in the liver. If the drug is completely soluble and permeable, not complexed or degraded or effluxed or metabolized in gut, and not metabolized or effluxed in liver, F will be close or equal to unity. Poor oral bioavailability of immunosuppressive drugs such as tacrolimus, sirolimus, and cyclosporine is attributed primarily to metabolism and efflux of these drugs in the gut wall.

Distribution

Distribution of a drug in the body is influenced by many factors. The apparent volume of distribution is a hypothetical volume which refers to volume of plasma required to occupy the total amount of drug in the body at the concentration observed in plasma. The volume of distribution is influenced by blood or plasma volume, total body water, and binding to plasma and tissue proteins. The volume of distribution is represented by Eq. (29.15). Immunosuppressive drugs such as tacrolimus, sirolimus, and cyclosporine show very large volume of distribution indicating extensive binding to different tissues and lower levels in circulation. In contrast, large molecules such as biologics and antibodies show a small volume of distribution and are predominantly restricted to plasma or blood.

Metabolism

Metabolism is the process of converting one chemical into another by removal of specific group or addition of a molecule or both to make it easily excretable. If a drug is eliminated only by metabolism, metabolic clearance accounts for total

body clearance. For a drug metabolized by the liver only, hepatic clearance will represent metabolic as well as total body clearance. Hepatic clearance CL_h of the drug can be influenced by hepatic blood flow Q, the intrinsic metabolic capacity of the liver for that drug CL_{int}, and the fraction unbound in plasma f_u as follows:

$$CL_h = \frac{Qf_uCL_{int}}{Q + f_u \times CL_{int}}(mL/min) \tag{29.23}$$

For high hepatic intrinsic clearance drugs, the equation transforms to

$$CL_h \approx Q(mL/min) \tag{29.24}$$

indicating that the highest clearance of the drug by the liver corresponds to blood flow to the liver and changes in the blood flow to liver significantly change hepatic clearance. For low intrinsic clearance drugs, the equation simplifies to

$$CL_h = f_u \times CL_{int}(mL/min) \tag{29.25}$$

indicating that changes in f_u and CL_{int} will significantly influence the hepatic clearance and changes in hepatic blood flow may have limited to no effect on the hepatic clearance of the drug.

Hepatic blood flow, intrinsic clearance, and unbound fraction in plasma can be altered by physiological, pathological, and pharmacological factors. Treatment with enzyme inducers such as rifampin, phenobarbital, phenytoin, and carbamazepine increases intrinsic clearance, whereas inhibitors such as ketoconazole, ritonavir decrease intrinsic clearance. Several drug−drug interactions have been reported with cytochrome P450 3A4 enzyme substrate immunosuppressive drugs such as tacrolimus, sirolimus, and cyclosporine with both inducers and inhibitors of these enzymes warranting extreme caution while taking multiple medications. Altered physiological states such as liver disease can lead to decreased albumin concentrations and decreased plasma protein binding leading to increased free fraction and clearance. Subsequent to liver transplantation the increase in albumin levels in patients decreases unbound fraction of the drugs such as mycophenolic acid in plasma resulting in decreased overall total body clearance.

Clearance of the drug can be influenced by age, gender, diet, co-medications, and environmental and genetic factors. It is well known that clearance of immunosuppressants such as tacrolimus, sirolimus, cyclosporine, and mycophenolic acid is higher in pediatric patients than in adults requiring higher dose on a milligram/kg basis to achieve similar exposure and clinical outcomes.

For a drug metabolized by liver, the fraction of bioavailable drug after first pass can be calculated using the equation:

$$F = 1 - E \tag{29.26}$$

where E is the fraction of the drug extracted in the liver in a single passage, by incorporating Eq. (29.9) for E, Eq. (29.20) transforms to

$$F = 1 - \frac{CL}{Q} \tag{29.27}$$

Substituting Eq. (29.12) for CL;

$$F = 1 - \frac{Dose_{IV}}{AUC_{IV} \times Q} \tag{29.28}$$

Therefore, with information on IV dose and corresponding AUC and assuming physiologic liver blood flow of 1.5 L/min, theoretical bioavailability of the drug can be predicted after IV administration and the deviation from the predicted values can be attributed to incomplete absorption, degradation of the drug in gut or metabolism or efflux for over prediction, and saturable hepatic metabolism for under prediction.

Further modifying the above equation by incorporating Eq. (29.17) into Eq. (29.21), F can be related to blood flow, intrinsic clearance, and f_u as follows:

$$F = \frac{Q}{Q + f_u \times CL_{int}} \tag{29.29}$$

For very low clearance drugs (low intrinsic CL_{int}), F approaches 1. Whereas for drugs with high intrinsic clearance, the equation transforms to

$$F \approx \frac{Q}{f_u \times CL_{int}} \tag{29.30}$$

Increase in blood flow increases F, increase in F_u and CL_{int} decreases F.

Excretion

Excretion is irreversible removal of drug from the body. Kidneys are the primary organs for elimination of unchanged drug from body. Renal clearance CL_r of the drug can be estimated by the following equation:

$$CL_r = \frac{\text{Cumulative amount excreted in urine}}{AUC_{Plasma(o-\infty)}} \tag{29.31}$$

or

$$CL_r = \frac{\text{Amount excreted in urine over a time period}}{\text{Plasma concentration of drug at mid point of urine collection}}$$

The above equation can be used to calculate CL_r of the drug whether the drug was given by intravascular or extravascular route.

A drug is excreted through kidneys by glomerular filtration and/or active secretion and may undergo tubular reabsorption. The renal clearance can be represented by the following equation:

$$CL_r = (CL_{filtration} + CL_{secretion})(1 - F_{Reabsorption}) \tag{29.32}$$

where $CL_{filtration}$ is the filtration clearance, $CL_{secretion}$ is the secretion clearance, and $F_{Reabsorption}$ is the fraction reabsorbed from the tubules.

Certain drugs are excreted in the bile. If the unchanged drug or a metabolite of drug is excreted in bile, and this metabolite can be converted into the parent drug in gut and absorbed back into systemic circulation leading to a process of enterohepatic circulation, which leads to increase exposure of the drug in patient. Mycophenolic acid, an immunosuppressant, and indomethacin, an NSAID, undergo conjugation and the conjugates secreted in bile are cleaved in gut and reabsorbed back into systemic circulation.

OBJECTIVE 29.3. OUTLINE PHARMACOMETRICS

Pharmacometrics is the science of understanding and mathematically explaining the relationship between dose–concentration effects with the application of pharmacology, biology, mathematics, and biostatistics principles. In general term, pharmacometrics encompasses modeling and simulation for PK, pharmacodynamics (PD), and disease progression. Pharmaceutical and Biotech companies and regulatory agencies all over the world have invested heavily in the development of proper expertise to analyze data generated during drug development process from preclinical to clinical stages to understand drug exposure and disease progression for better efficacy with minimum toxicities. The purpose of this segment of the book chapter is to provide a general idea about types of modeling and simulation and their application in real-world research.

PKPD modeling and simulation could be divided into two major categories: (1) population pharmacokinetic modeling and simulation (PopPK) and (2) physiologically based pharmacokinetic modeling and simulation (PBPK).

Population Pharmacokinetic Modeling and Simulation

In conventional PK, data are analyzed either by compartmental or noncompartmental approaches. In both cases, extensive samples are collected over a considerable period of time based on drug clearance, half-life, and the dosing interval. Then standard two-stage approach is applied to analyze the data. In the first stage, the PK parameters are calculated at individual level and in the second stage, individual PK parameters are combined together to estimate population parameters in addition to standard deviation (SD) or variance of the parameters in the population. However, this method has certain limitations; it typically uses a small number of subjects; the calculated SD or variance is based on the population studied

without accounting for the variability that may arise in a typical patient population. Another issue with conventional PK analysis is the need to collect a lot of samples after a given dose in each patient to estimate various PK parameters. In general, the intense sample collection leads to additional complications including need for withdrawal of large blood volume and increased cost of sample analysis. Moreover, intense sampling becomes inconvenient for some special population such as children and pregnant women. For these reasons, application of limited or sparse sampling strategy is becoming more popular in pediatric and pregnancy research, where sparse sampling implies the collection of smaller number of samples from each individual. The use of sparse or limited sampling to obtain population PK parameters was the main incentive behind development of population PK analysis approaches. In population pharmacokinetic approach, sparse or limited sampling could be used to estimate same PK parameters as we do in conventional PK analysis with extensive sampling from a large number of patients in the clinical setting. In therapeutic drug monitoring (TDM) programs, we only collect limited samples per dosing interval, and these data cannot be readily used to estimate PK parameters by conventional approach, but by applying PopPK approach, we can use those data to estimate PK parameters in patients who actually use the drug. Additionally, collecting less number of samples is convenient for the patients, and it reduces the time and cost involved in the drug research and development. Currently, various software are available for PopPK analysis such as NONMEM, MONOLIX, and Phoenix NLME, etc.

Application of Sparse or Limited Sampling in PopPK

Population PK modeling and simulation is applied in normal clinical practice to improve patient care. First, a population PK model is developed for a drug using a patient population, where typical values for the PK parameters of a drug are calculated. Several patient covariates are incorporated into the model and a final model is developed.[13,14] Once the model is developed, this developed model could be used to predict drug exposure and response in a different disease, different patient population, and with different dosage regimen. In practice, very limited numbers of samples are collected from each patient; measured plasma levels are given as an input into the developed population model to generate a model for individual patient. Finally, individualized model is used to predict drug exposure and effects relationship for different dosage regimen.

Practical Example

Vancomycin is a commonly used medication to treat gram-positive cocci (*Staphylococcus aureus* and *Staphylococcus epidermidis*) as well as in empirical antibiotic therapy in patients with malignant hematological diseases. However, PK of vancomycin shows a high interindividual variability, primarily associated with patents age and disease condition. Additionally, PK of vancomycin is different in oncology patients as compared with other patient populations. Very limited data are available about the variability in vancomycin PK in pediatric and oncology patients. A population pharmacokinetic model in pediatric patient was developed using patient's TDM samples and major covariates affecting the PK of vancomycin were identified. Then this developed model was used to calculate individual patient-tailored dose to achieve target plasma levels for better therapeutic effect. Following equation was used to calculate patient clearance, and resultant clearance value was used to calculate final individualized dose.

$$\text{Clearance} = 4.37 \times (\text{body weight} \div 20.7)^{0.67} \times (\text{creatinine clearance} \div 191)^{1.03}$$

where, 4.37 is the reference clearance, 20.7 is the reference body weight, and 191 is the reference creatinine clearance.

Traditional dose of vancomycin (60 mg/kg/day) results in only 15% of the children achieving target AUC/MIC. Whereas, based on PopPK simulated trial using the above-mentioned factors for vancomycin clearance calculation, children receiving a dose of 90 mg/kg/day have a risk of under dosing in only 20%. This example shows the importance of population PK modeling and simulation for individualized patient dosing.[15]

PHYSIOLOGICALLY BASED PHARMACOKINETIC MODELING AND SIMULATION

Pharmaceutical scientists have been using modeling and simulation technique for a long time. Quantitative structure activity relationship and quantitative structure property relationship models have been very useful in the drug development process. Unfortunately, the predictive powers of these modeling approaches are not always good, and their predictability is mainly depended on the compounds' characteristics that were used to develop the model. On the other hand, mechanistic models are predefined and can explain the underlying mechanism for variability in the PK parameters as well as can extend

the known to unknown for prediction. Physiologically based pharmacokinetic models integrate the physicochemical properties of a drug along with the drug-independent properties of biological system to provide a mechanistic representation of the compounds in a specific biological system, enabling prediction or simulation of the plasma concentration time profile. Additionally, PBPK provides physiological representation at the organ level. This modeling technique is very useful to predict drug concentration−time profile in an organ or tissue, where collection of real biological sample is practically impossible such as brain, fetus, and other organs.[16]

Generally, whole body PBPK model connects all the major organs such as heart, lung, kidney, brain, gut, pancreas, liver, skin, bone, and adipose tissue (Fig. 29.5) in the selected species. Each of the organs represents a single compartment in the PBPK model. These organs are linked by artery and vein in such a way that they represent their exact blood flow from in and out of an organs. In a PBPK model, we can populate drug-metabolizing enzymes, transporters, proteins, volume of the organ, and blood flow for each of the organs to predict the impact of various disease states or different patient populations or drug−drug interactions.

Application of PBPK modeling and simulation: PBPK modeling is very useful in drug development and research. Due to the improvement of computational power, companies are developing robust PBPK modeling software with advanced applications and features. Mostly known PBPK software include Simcyp, PKSIM, Gastroplus, ADMEWORKS,

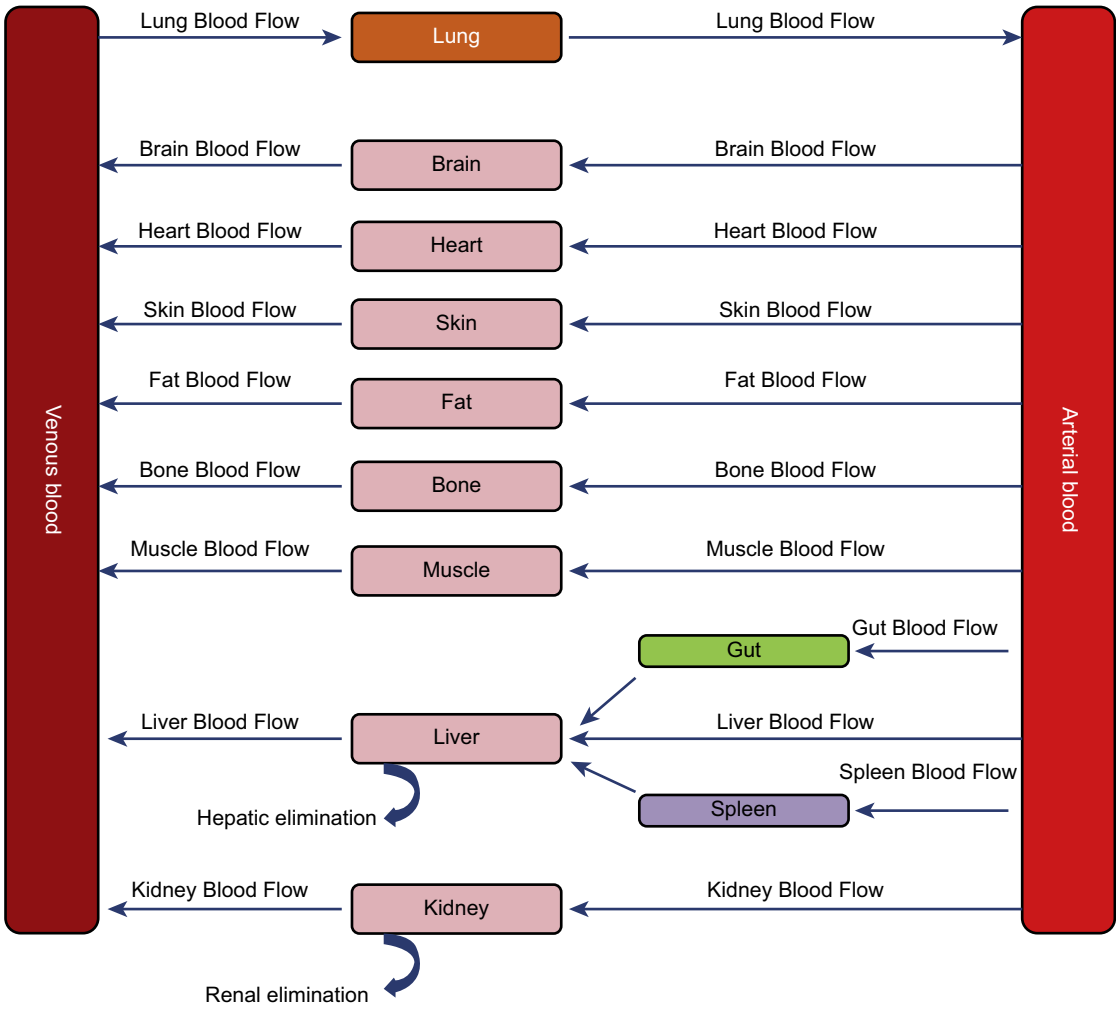


FIGURE 29.5 Compartments in whole body physiologically based pharmacokinetic modeling and simulation model. *Representation of whole body PBPK model. Organs are connected with corresponding arterial and venous blood flow; liver and kidney are the two major drug-eliminating organs.*

DDI simulators, etc can be usedto simulate pharmacokinetic profiles in virtual patients, to predict possible drug−drug interaction, extrapolate data from preclinical models to human, simulate human plasma concentration time profile from in vitro data, predict drug exposure from one patient population to another.

CONCLUSION

In summary, pharmacokinetic parameters are very important to estimate how much drug has to be administered to achieve a desired drug exposure for optimal therapeutic effects, avoiding unwanted side effects and/or therapeutic failure. Additionally, pharmacokinetic parameters help us to understand the mechanisms responsible for interpersonal and/or intrapersonal variability in drug exposure. Pharmacometrics is an emerging field in the area of PK, where mathematical modeling is used to describe/interpret pharmacokinetic data.